

Tutorial set 1

1. Find the inverse of the matrices, if possible

$$\begin{pmatrix} 5 & 3 \\ 4 & 2 \end{pmatrix} \quad \begin{pmatrix} 2 & -3 \\ 1 & 3 \end{pmatrix} \quad \begin{pmatrix} -2 & 6 \\ 3 & -9 \end{pmatrix}$$

2. Determine whether or not each of the following matrices is symmetric, that is $A^T = A$ or skew-symmetric $A^T = -A$:

$$\begin{pmatrix} 5 & -7 & 1 \\ -7 & 8 & 2 \\ 1 & 2 & -4 \end{pmatrix} \quad \begin{pmatrix} 0 & 4 & -3 \\ -4 & 0 & 5 \\ 3 & -5 & 0 \end{pmatrix}$$

3. Apply the Gram-Schmidt orthogonalization process to find an orthogonal basis and then an orthonormal basis for the space spanned by

$$v_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix} \quad \begin{pmatrix} 1 \\ 2 \\ 4 \\ 5 \end{pmatrix} \quad \begin{pmatrix} 1 \\ -3 \\ -4 \\ -2 \end{pmatrix}$$

4. Find the eigenvalues and normalized eigenvectors of the matrix

$$\Omega = \begin{pmatrix} 1 & 3 & 1 \\ 0 & 2 & 0 \\ 0 & 1 & 4 \end{pmatrix}$$

Is it Hermitian ? Are the eigenvectors orthogonal ?

5. Consider the matrix

$$\Omega = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}$$

Is it Hermitian ? Find its eigenvalues and eigenvectors. Show that $U^\dagger \Omega U$ is diagonal, U being the matrix of eigenvectors of Ω .

6. Consider

$$\Omega = \frac{1}{2} \begin{pmatrix} 2 & 0 & 0 \\ 0 & 3 & -1 \\ 0 & -1 & 3 \end{pmatrix}$$

(i) Show that the eigenvalues are 1, 1 and 2. (ii) Show that $|\omega = 2\rangle$ is any vector of the form

$$\frac{1}{\sqrt{2a^2}} \begin{pmatrix} 0 \\ a \\ -a \end{pmatrix}$$

(iii) Show that $\omega = 1$ eigenspace contains all vectors of the form

$$\frac{1}{\sqrt{b^2 + 2c^2}} \begin{pmatrix} b \\ c \\ c \end{pmatrix}$$

either by feeding $\omega = 1$ into the equations or by requiring that the $\omega = 1$ eigenspace be orthogonal to $|\omega = 2\rangle$.